Cairo University

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Homework 1

ELC 4028 – Artificial Neural Networks

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1 Perceptron Learning Rule Derivation

In this section, we derive the perceptron learning rule step by step starting from the model definition.

1.1 Perceptron Model

The perceptron computes a linear function of the input:

$$a = \mathbf{w}^T\mathbf{x} + b$$

The predicted label is:

$$\hat{y} = \text{sign}(a)$$

Where:

- $\mathbf{x}$: input feature vector
- $\mathbf{w}$: weight vector
- b: bias term
- a: activation
- $\hat{y}$: predicted label
- $y \in \{+1, -1\}$: true label

1.2 Correct vs Incorrect Prediction

The prediction is correct if:

$$y(\mathbf{w}^T\mathbf{x} + b) > 0$$

and incorrect if:

$$y(\mathbf{w}^T\mathbf{x} + b) \leq 0$$

Explanation:

- If $y = +1$, we need $(\mathbf{w}^T\mathbf{x} + b) > 0$
- If $y = -1$, we need $(\mathbf{w}^T\mathbf{x} + b) < 0$

Multiplying by y ensures the expression is positive only when the prediction is correct.

1.3 Objective

If a prediction is incorrect, we aim to:

- Update $\mathbf{w}$ and b
- Move the decision boundary toward correct classification

1.4 Proposed Update Rule

We propose the following update:

$$\mathbf{w}_{new} = \mathbf{w} + \eta y \mathbf{x}$$

$$b_{new} = b + \eta y$$

where $\eta > 0$ is the learning rate.

Update condition:

$$y(\mathbf{w}^T\mathbf{x} + b) \leq 0$$

1.5 Why the Update Works

We analyze the quantity:

$$y(\mathbf{w}_{new}^T\mathbf{x} + b_{new})$$

Substitute the updated parameters:

$$= y\left((\mathbf{w} + \eta y \mathbf{x})^T\mathbf{x} + (b + \eta y)\right)$$

$$= y\left(\mathbf{w}^T\mathbf{x} + \eta y \mathbf{x}^T\mathbf{x} + b + \eta y\right)$$

$$= y(\mathbf{w}^T\mathbf{x} + b) + \eta y^2(\mathbf{x}^T\mathbf{x} + 1)$$

Since:

$$y^2 = 1$$

we obtain:

$$= y(\mathbf{w}^T\mathbf{x} + b) + \eta(\|\mathbf{x}\|^2 + 1)$$

Observation:

- $\|\mathbf{x}\|^2 + 1 > 0$
- Therefore, the value strictly increases after the update

Conclusion:

The update improves the classification score and moves the prediction toward the correct class.

1.6 Final Learning Rule

Weight update:

$$\mathbf{w} \leftarrow \mathbf{w} + \eta y \mathbf{x}$$

Bias update:

$$b \leftarrow b + \eta y$$

Apply only when:

$$y(\mathbf{w}^T \mathbf{x} + b) \leq 0$$

1.7 Interpretation

The perceptron learning rule:

- Updates parameters only when an error occurs
- Moves the decision boundary in the correct direction
- Reinforces correct classification over time

This rule forms the foundation of linear classifiers and connects directly to margin-based methods such as Support Vector Machines.